ALEXANDRU CRISTIAN CALIN

alexandrucalin12@gmail.com | +44 7789 946864 | UK settled status | London, UK

EDUCATION

London School of Economics Sep 2024 to Sep 2025

MSc Applicable Mathematics. Modules: machine learning and data mining, game theory, algorithms.

Imperial College London Jul 2024 to Aug 2024

Summer Research Student, Machine Learning and Quantitative Finance. Built a time series model in R to forecast energy commodity prices and presented the results and scenario implications to faculty and the cohort.

University of Manchester Sep 2020 to Jun 2024

BSc Mathematics. Modules in probability theory, statistical inference, numerical analysis and time series analysis.

EXPERIENCE

TikTok (ByteDance) Sep 2025 to Present

Strategy and Operations Manager, TikTok Live, London

- Researched and compared four European markets to judge where growth potential was highest.
- Built SQL and Python segmentation of the portfolio of 55 partner businesses to rank them by revenue potential and find the highest value segments.
- Presented market analysis, partner acquisition strategy and a return on investment comparison of acquisition channels to European leadership to guide resource allocation.
- Designed incentive experiments, measured them against a baseline and dropped three that did not change partner behaviour, refocusing effort on what worked.
- Built AI workflows with large language models that analysed partner data and drafted first versions of growth strategies, checking and correcting outputs before use.
- Reviewed content and campaign performance with partners and turned it into clear, practical plans with success metrics.

Regnum Aegyssus Jun 2025 to Sep 2025

Financial Analyst, Romania

- Modelled volatility transmission across Eastern European equity markets in Python using rolling window multivariate estimation; the research is under journal review.
- Turned complex findings into clear written conclusions for supervisors, explaining assumptions and limits.

RESEARCH AND PROJECTS

Do AI Agents Play Like People? (independent project, in progress) Oct 2026 to Present

- Comparing how humans, bounded rationality models, reinforcement learning agents and open weight language models play the same 2,416 two player games, using a public dataset of 93,460 human decisions (Zhu et al., 2025).
- Developing the evaluation in Python, scoring each model by how well it predicts human choices and where it fails.

Volatility Spillovers in Central European Markets (SSRN working paper)

- Combined desk research and data analysis in R to show how volatility spread between the German, Polish, Hungarian and Czech stock markets after the war in Ukraine began.

Credit Default Prediction Model (LSE) May 2025 to Jun 2025

- Compared neural networks, k nearest neighbours and random forests on 30,000 credit card records, reaching an AUC-ROC of 0.82 through feature engineering and hyperparameter tuning.
- Used feature importance to explain the main drivers of default in plain, actionable terms.

Multiagent Reinforcement Learning in Energy Markets (LSE MSc dissertation, SSRN) 2025

- Planned and delivered an independent research project, modelling in Python how participants with different strategies shape outcomes for a whole market.

SKILLS

- **Data and analysis:** Python (pandas, NumPy, scikit learn, statsmodels, matplotlib), SQL, R, Excel; segmentation, experiment design, time series analysis.
- **Research and storytelling:** market and opportunity assessment, desk research, executive presentations and reports, explaining quantitative findings to non technical audiences.
- **AI:** building workflows with large language models, prompt design, reviewing AI outputs for errors and hidden assumptions.
- **Languages:** English (fluent), Romanian (native).